Diabetes Prediction Using an Artificial Neural Network with a Calibrated, Explainable Ensemble

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Problem and Objectives

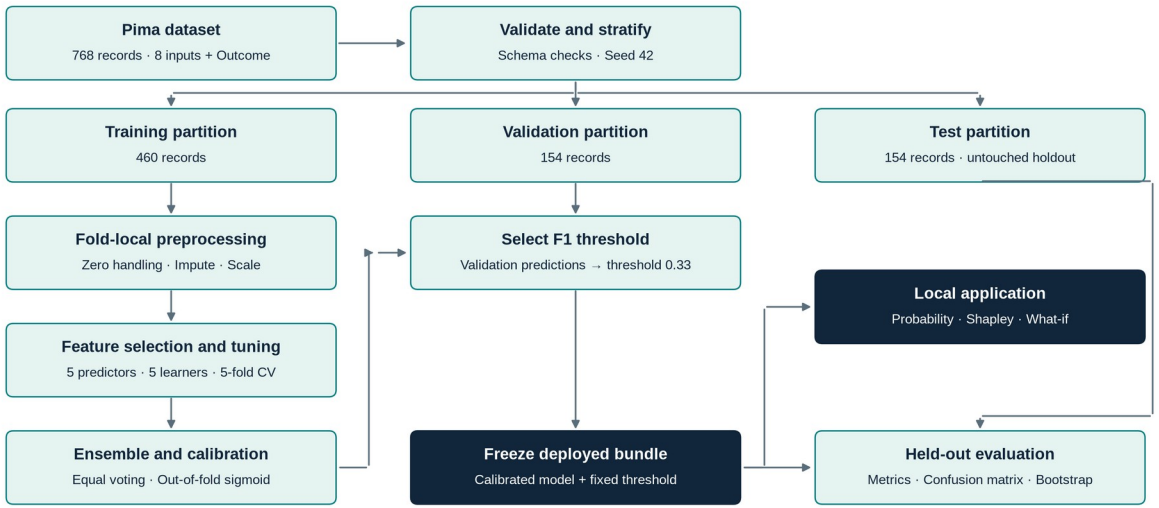
- Binary classification of the Pima Indians Diabetes Outcome label from 8 measurements (768 women, 35% positive)
- Not a diagnosis or onset forecast: educational demonstrator
- Objectives: leakage-controlled pipeline, ANN vs 4 learners, calibrated ensemble, validated threshold, ablation, explanations, local app

Literature Survey and Gap

- Sisodia 2018: Naive Bayes 76.3% on Pima (10-fold CV)
- Zou 2018: RF, NN with PCA/mRMR; Sneha 2019: feature selection helps
- Tasin 2023, Hasan 2025: ensembles/AutoML with SHAP and apps
- Grinsztajn 2022: trees often beat deep nets on tabular data
- Gap: single accuracy numbers, little calibration reporting, explanations of stand-alone models, optimistic CV

Proposed Methodology

- Fixed-seed stratified split: 460 train / 154 validation / 154 test
- Zeros in Glucose, BP, Skin, Insulin, BMI treated as missing; imputation, scaling, selection fitted per fold
- Five tuned models (6 random configs x 5-fold CV each)
- Equal-weight soft voting, out-of-fold sigmoid calibration
- Threshold 0.33 chosen on validation (max F1); test used once



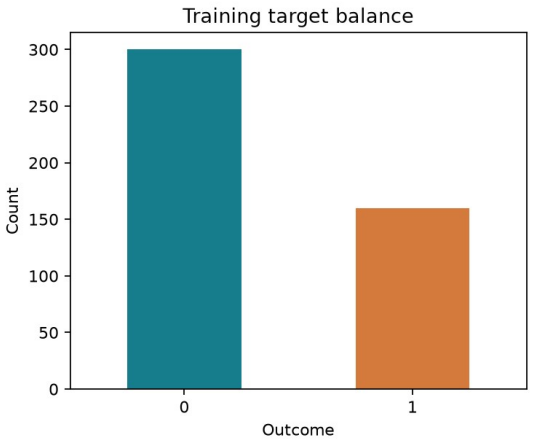
Training fits transformations and models; validation chooses the threshold; test data only evaluates the frozen model.

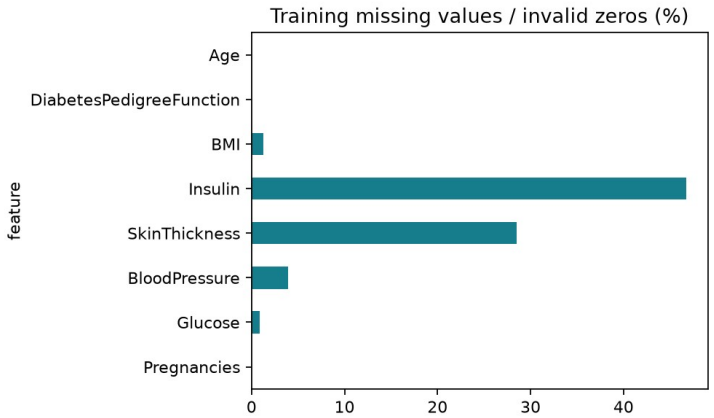
ANN Architecture

- Input: 5 selected features
- Hidden: 32 ReLU units: $h = \mathrm{ReLU}(W_1 x + b_1)$
- Output: $p = \mathrm{sigmoid}(W_2 h + b_2)$
- Loss: L2-regularised cross-entropy; Adam, lr 0.003, alpha 0.001, early stopping
- Selected CV AUC 0.821

Dataset and Exploration

- Insulin 46.7% and skin thickness 28.5% missing (zeros)
- Correlation with Outcome: Glucose 0.50, BMI 0.34, Insulin 0.31
- Dummy classifier: 64.9% accuracy, recall 0



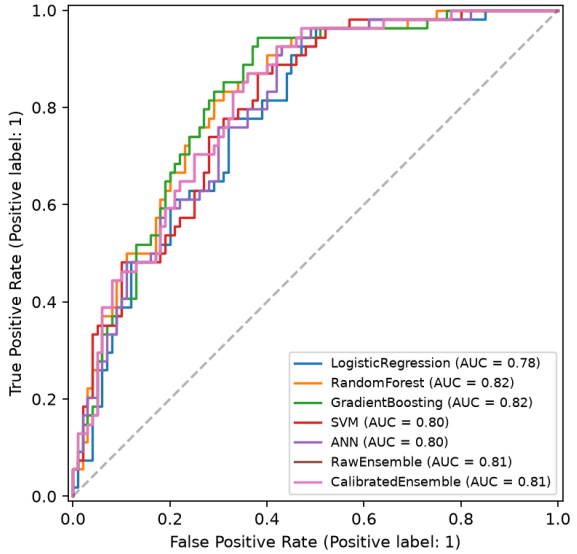


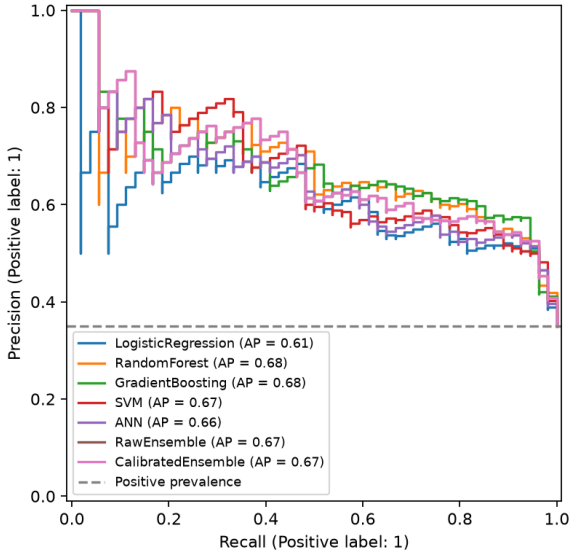
Feature Representation Ablation (5-fold CV AUC)

- Logistic: original 0.826, engineered 0.825, selected 0.821
- ANN: original 0.795, engineered 0.756 (unstable), selected 0.808
- Selected 5 features chosen on training data only

ROC and Precision-Recall

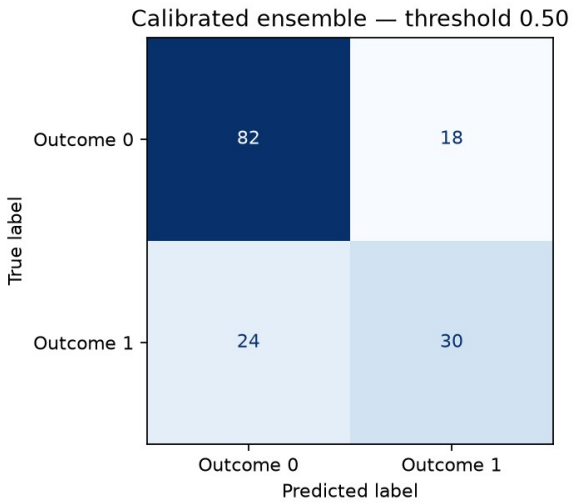
- All models far above no-skill
- Gradient boosting AUC 0.822, RF 0.821, ensemble 0.811, SVM 0.800, ANN 0.796, LR 0.783

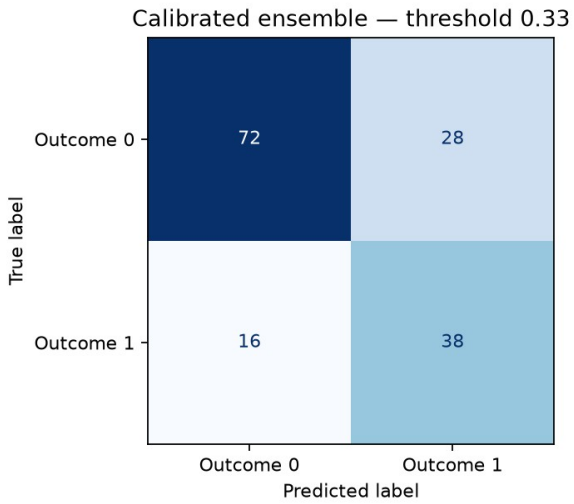




Deployed Ensemble Results (test, n = 154)

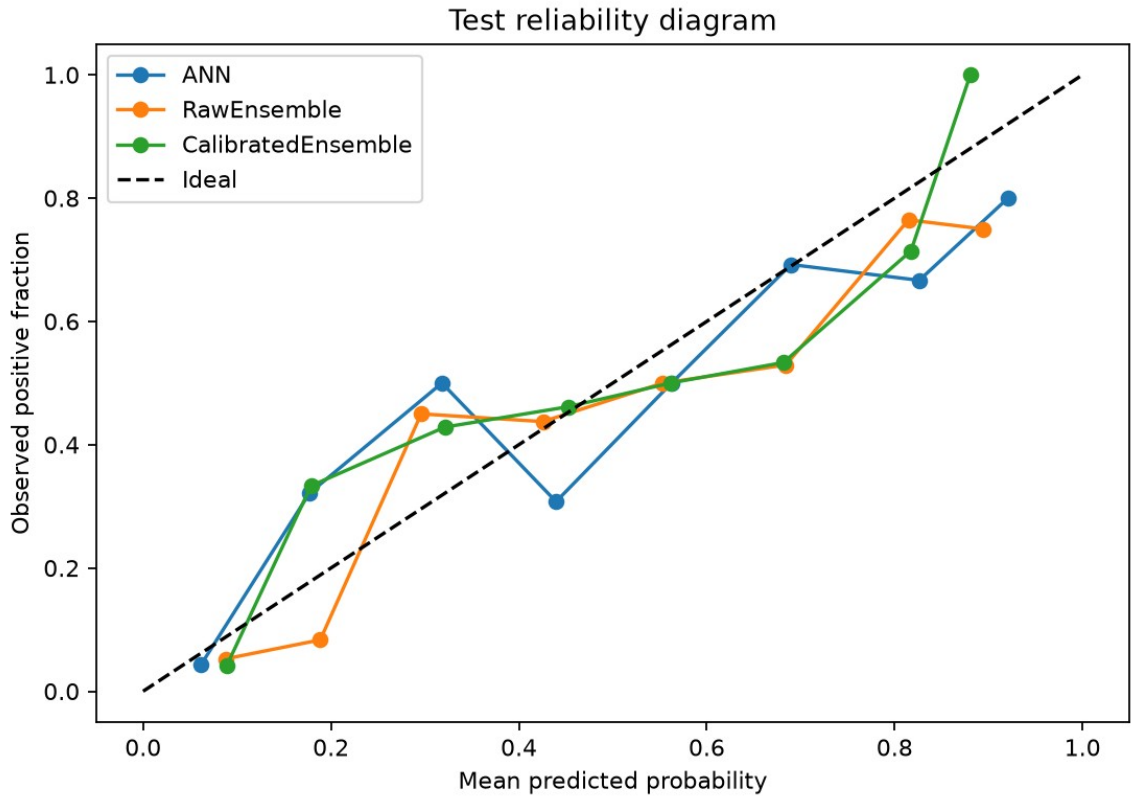
- AUC 0.811 (95% CI 0.736-0.876)
- At threshold 0.33: accuracy 71.4%, recall 70.4%, specificity 72.0%, F1 0.633
- Confusion: TN 72, FP 28, FN 16, TP 38
- At 0.50: recall only 55.6%: threshold matters
- McNemar vs ANN p = 1.0; AUC difference +0.015 (CI 0.0006-0.032)





Calibration

- ECE 0.125 -> 0.086 (better)
- Brier 0.1709 -> 0.1746 (slightly worse)
- AUC unchanged (monotonic mapping)
- Mixed evidence, reported honestly

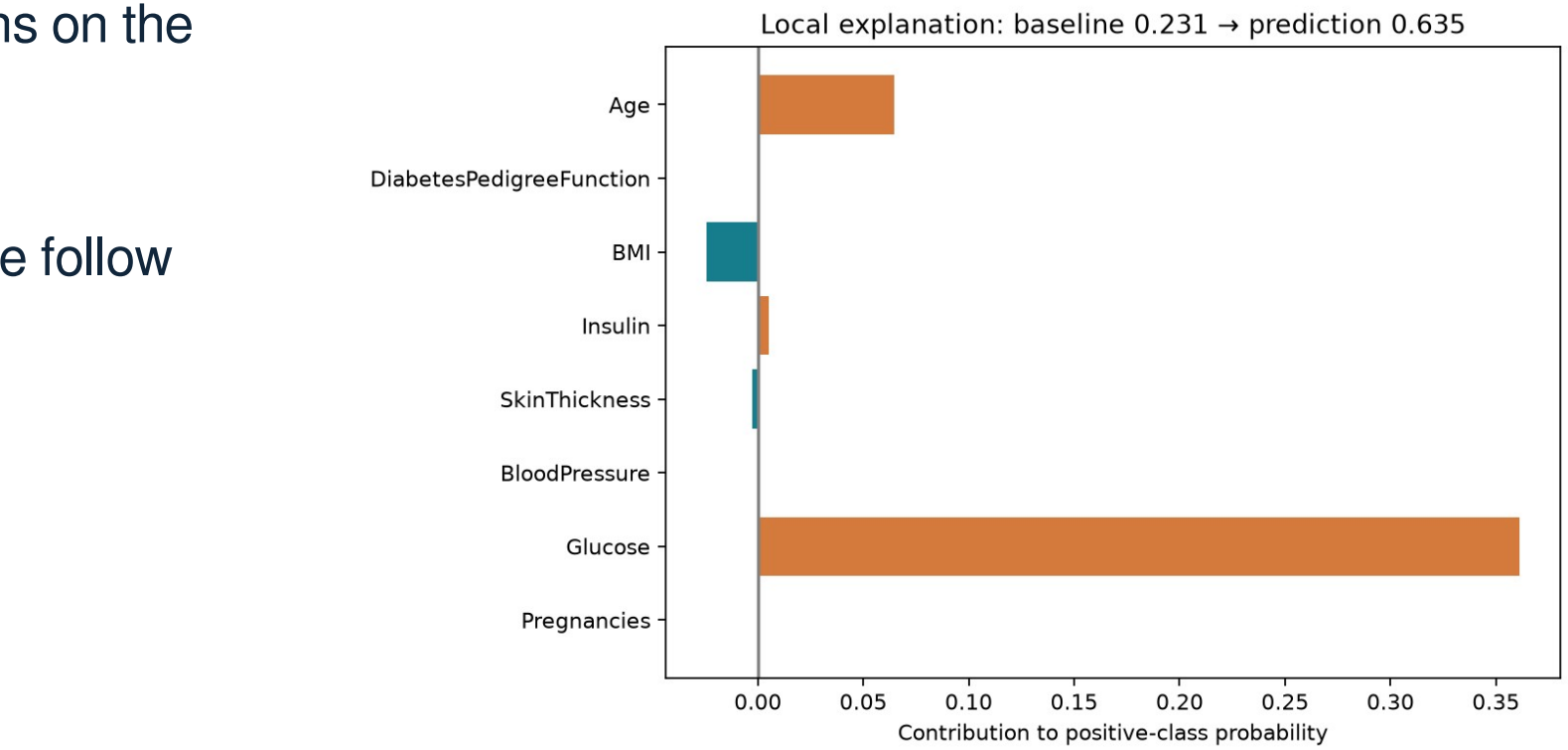


Ablation Study

- Representation: engineered features hurt the ANN
- Models: dummy 0.50 < LR 0.783 < ANN 0.796 < SVM 0.800 < ensemble 0.811 < RF 0.821 < GB 0.822 (test AUC)
- Calibration: ECE -31%, Brier +0.004
- Threshold: recall 55.6% -> 70.4% for -10 pts specificity
- Permutation: Glucose -0.142 AUC, BMI -0.041, Age -0.036

Explainability

- Exact Shapley over 256 coalitions on the full deployed pipeline
- Additivity error 2.2e-16
- Glucose dominates; BMI and age follow



State-of-the-Art Comparison

- Literature: ~76% (ADAP, Naive Bayes), 75.7% ANN, 81.8% boosting (protocols differ)
- This work: 71.4-74.0% on a strict one-shot held-out test
- Lower numbers reflect stricter protocol; adds calibration, validated threshold, CI and explanations

Limitations and Future Work

- 154-row test set: wide intervals
- Heavy missingness; single historical population; no external validation
- Non-nested CV is optimistic
- Future: nested CV, external cohort, multiple imputation, subgroup analysis

Conclusion

- ANN competitive but not superior on small tabular data
- Calibrated ensemble: AUC 0.811, recall 70.4% at validated threshold
- Glucose is the dominant predictor
- Educational demonstrator, not a clinical tool

References (selected)

- Smith et al. 1988, ADAP, SCAMC
- Sisodia & Sisodia 2018, Procedia CS 132
- Zou et al. 2018, Front. Genet. 9:515
- Lundberg & Lee 2017, NeurIPS; Guo et al. 2017, ICML
- Breiman 2001; Friedman 2001; Kingma & Ba 2015
- Full list of 35 references in the report